12.16: $\mu_0 \cdot \cancel{2\pi} \frac{r^2}{R^2} \cdot I = B \cdot 2\pi r$

则 $B = \frac{\mu_0 r I}{2\pi R^2}$

$d\Phi = B dS$

$= \frac{\mu_0 r I}{2\pi R^2} \cdot dr$

$\Phi = \int_0^R \frac{\mu_0 I}{2\pi R^2} \cdot r dr = \frac{\mu_0 I}{2\pi R^2} \cdot \frac{1}{2} R^2 = \frac{\mu_0 I}{4\pi} = 10^{-6} Wb$

12.17 (1) $B \cdot 2\pi r = \frac{I \cdot \pi r^2}{\pi a^2} \cdot \mu_0$

则 $B = \frac{\mu_0 I r}{a^2 \cdot 2\pi}$ $(r<a)$

(2) $a<r<b$

$B \cdot 2\pi r = I \mu_0$

$B = \frac{I \mu_0}{2\pi r}$ $(a<r<b)$

(3). $b<r<c$

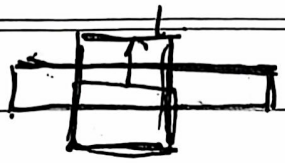
$I' = I - \frac{I(r^2-b^2)\pi}{\pi(c^2-b^2)} = I - \frac{r^2-b^2}{c^2-b^2} I = \frac{c^2-r^2}{c^2-b^2} I$

$B \cdot 2\pi r = \mu_0 \cdot I'$

$B = \frac{\mu_0 (c^2-r^2) I}{2\pi r (c^2-b^2)}$

(4). $r>c$. $B=0$

12.20



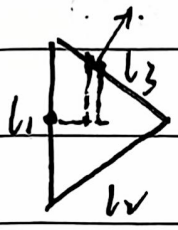
$2B\cdot l = j_2\cdot l\cdot \mu_0$，则 $B=\frac{j_2\mu_0}{2}$.

则 P: $B=\frac{\mu_0}{2}j_2-\frac{\mu_0}{2}j_1$

Q: $B=\frac{\mu_0}{2}j_2+\frac{\mu_0}{2}j_1$

12.26

$B\cdot 2\pi r=\mu_0 I_1$. $B=\frac{\mu_0 I_1}{2\pi r}$, $r=b-\frac{\sqrt{3}}{6}a$



$$F_{l_1}=a\cdot\frac{\mu_0 I_1}{2\pi r}\cdot I_2=\frac{\mu_0 I_1 I_2 a}{2\pi(b-\frac{\sqrt{3}}{6}a)}.$$

$$dF_{l_2}=\frac{\mu_0 I_1}{2\pi r}I_2\cdot dr\cdot\frac{2}{\sqrt{3}}\cdot\frac{1}{2}=\frac{\mu_0 I_1 I_2}{2\sqrt{3}\pi}\cdot\frac{dr}{r}.$$

$$F_{l_2}=\int_{b-\frac{\sqrt{3}}{6}a}^{b+\frac{\sqrt{3}}{3}a}dF_{l_2}=\frac{\mu_0 I_1 I_2}{2\sqrt{3}\pi}\ln\frac{b+\frac{\sqrt{3}}{3}a}{b-\frac{\sqrt{3}}{6}a}.$$

故 $F_{总}=2F_{l_2}-F_{l_1}=\frac{-\mu_0 I_1 I_2 a}{2\pi(b-\frac{\sqrt{3}}{6}a)}+\frac{\sqrt{3}\mu_0 I_1 I_2}{3\pi}\ln\frac{b+\frac{\sqrt{3}}{3}a}{b-\frac{\sqrt{3}}{6}a}$

12.28 $mg\sin\theta\cdot R=NBIL\times 2\times R\sin\theta$.

$$I=\frac{mg\sin\theta R}{NBL\cdot 2R\sin\theta}=\frac{0.25\times 9.8}{10\times 0.1\times 2\times 0.5}=2.45A$$

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12.30. (1) $B=\frac{\mu_0 I R^2}{2(R^2+x^2)^{\frac{3}{2}}}$，$x\to 0$，则 $B=\frac{\mu_0 I}{2R}$.

$$B_b=\frac{\mu_0 I}{2b}\times\frac{1}{2}=\frac{\mu_0 I}{4b}$$

$$B_a=\frac{\mu_0 I}{4a}$$

则 $B=B_a+B_b=\frac{\mu_0 I}{4a}+\frac{\mu_0 I}{4b}$，垂直纸面向内

(2). $\vec{p_m}=NIS\vec{e_n}$

$$=I\cdot\frac{\pi(a^2+b^2)}{2}$$

12.31. $dq=\frac{q}{\pi(R_2^2-R_1^2)}\cdot 2\pi r\,dr$

$f=\frac{\omega}{2\pi}$，则 $dI=f\cdot dq=\frac{q\cdot 2\pi r\,dr}{R_2^2-R_1^2}\cdot\frac{\omega}{2\pi}=\frac{\omega q r\,dr}{\pi(R_2^2-R_1^2)}$

$dM=dp_m\times B$

$=BI\cdot\pi r^2\cdot\sin\alpha$

$=B\cdot\frac{\omega q r\,dr}{\pi(R_2^2-R_1^2)}\times\pi r^2\sin\alpha$

$=B\cdot\frac{\omega q\sin\alpha}{R_2^2-R_1^2}\cdot\frac{1}{4}(R_2^4-R_1^4)$

$=\frac{B\omega q\sin\alpha}{4}(R_2^2+R_1^2)$

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(1). n型 (2).

12.38 $\Delta U = \frac{1}{nq} \cdot \frac{IB}{d}$.

$$n = \frac{IB}{\Delta U d q} = \frac{0.3T \times 1\times10^{-3}A}{6.55\times10^{-3}A \times 0.1\times10^{-2}m \times 1.6\times10^{-19}C}$$

$$= 2.86\times10^{20}\ 1/m^3$$

13.1 $B = \mu_0 n I$

$$I = \frac{4.2\times10^{-4}T}{4\pi\times10^{-7}\times1000} = 0.334A$$

$H\cdot 2\pi r = 2\pi r\cdot 1000\cdot 0.334A$

则 $H = 334A/m$.

$B = \mu_0\mu_r H$

$= 4\pi\times10^{-7}\times5\times10^{3}\times334$

$= 2.1T$

13.2 (1) $I = H_c \cdot R_{内} \times 2\pi$.

则 $I = \frac{500}{\pi} A/m \times 0.5\times10^{-3}m \times 2\pi$.

$= 0.5A$.

(2). $I = H_c R_{外} \times 2\pi = 0.8A$

~~13.4. (1) $\Phi = B\cdot S$ $B = \frac{2\times10^{-6}\mu_0 I}{\left(\frac{0.2}{2\pi}\right)^2 \times A}$~~

~~$B = \mu_0(1+\chi_m)H$. 则 $H = \frac{B}{\mu_0}$~~

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13.4. (1). $H \cdot l = I \times N$

$$H = \frac{0.032A \times 300}{0.3m} = 32\ A/m.$$

$$B = \frac{\Phi}{S} = \frac{2\times10^{-6}Wb}{1\times10^{-4}m^2} = 2\times10^{-2}\ T.$$

$$M = -H + \frac{B}{\mu_0} = \frac{2\times10^{-2}T}{4\pi\times10^{-7}} - 32A/m$$

$$= 1.59\times10^{4}\ A/m.$$

(2). $j_m = M = 1.59\times10^{4}\ A/m.$

(3). $\mu_r = \frac{B}{\mu_0 H} = \frac{2\times10^{-2}T}{4\pi\times10^{-7}\times 32A/m} = 497.$

13.8 (1). $N = \frac{V\cdot\rho}{\frac{M}{N_A}} = \frac{5\times10^{-2}m\times1\times10^{-4}m^2\times7.8\times10^{3}kg/m^3}{\frac{55.85\times10^{-3}kg}{6.02\times10^{23}mol^{-1}}}$

$$= 4.2\times10^{23}$$

$$P_m = N\cdot p_m = 4.2\times10^{23}\times1.8\times10^{-23}A\cdot m^2$$

$$= 7.57\ A\cdot m^2$$

(2) $M_y = M = BP_m$

$$= 7.57A\cdot m^2\times1.5T = 11.36\ N\cdot m.$$

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